PRACTITIONER KNOWLEDGE REPOSITORY

MODULE 5

ISDA, Netting & Collateral

The Legal Architecture of Derivative Markets

Blueprint Analogy: A derivative contract without an ISDA Master is like a building without foundations. The Master is the foundation; the Schedule, the walls; the CSA, the roof — load-bearing only if the layer beneath exists.

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Series Trail — Where We Are

M0	Greeks & Sensitivities
M1	Basel Accord Evolution
M2	FRTB
M3	CVA / DVA
M4	FVA / MVA
M5	**ISDA, Netting & Collateral ◀ THIS MODULE**
M6	KVA, ColVA & Capital Costs
MD	Practitioner's Dictionary

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OVERVIEW

Module 5 — What We Cover

1 **Why ISDA Exists**
Pre-netting world, 1987 catalyst, founding mandate

2 **The ISDA Document Hierarchy**
Master + Schedule + CSA — dependency rules & annotated architecture

3 **ISDA Evolution: 1992 → 2002 → Current**
Lehman stress test, Close-out Amount redesign, protocols

4 **Netting Mechanics**
Payment, close-out, portfolio netting — enforceability & capital impact

5 **Variation Margin**
Daily MtM settlement, eligible assets, haircuts, disputes

6 **Initial Margin & UMR**
MPoR buffer, SIMM v2.8, segregation, UMR phase-in

7 **Rehypothecation**
VM vs IM — rights, risks, UMR prohibition

8 **Repo & Securities Lending**
First-principles treatment — GC, special, haircuts, transformation, FVA link

9 **Regulatory & Legal**
EMIR, Dodd-Frank, MAS/HKMA, Hammersmith, Lehman, Archegos

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The Pre-Netting World — What ISDA Solved

✗ Cherry-Picking

Defaulting administrators selectively enforced profitable contracts, disclaimed losing ones. No mechanism to stop it.

✗ Gross Capital

Offsetting positions counted independently. Capital requirements massively overstated vs net economic risk.

✗ Bespoke Docs

Every bilateral trade separately documented. Inconsistent, slow, expensive, legally uncertain in stress.

ISDA SOLUTION

✓ **Close-out netting eliminates cherry-picking**

✓ **Net exposure for capital via SA-CCR**

✓ **Standardised docs reduce cost & uncertainty**

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The ISDA Three-Layer Stack

LAYER 1 — ISDA Master Agreement + Schedule

Inseparable — the Schedule is legally part of the Master Agreement

CAN stand alone ✓

▼ optional dependency

LAYER 2 — Credit Support Annex (CSA)

Governs VM and IM posting — CANNOT exist without a Master Agreement

CANNOT stand alone ✗

▼ governed by layers above

LAYER 3 — Trade Confirmations

Economic terms of each trade — governed by whichever Master and CSA applies

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Key Clauses by Layer

Layer	Clause Area	Key Substance
Master	Section 5 — Events of Default	8 enumerated triggers: Failure to Pay, Breach, Cross Default, Bankruptcy, Merger... Entitles non-defaulting party to terminate
Master	Section 6 — Close-out Netting	All trades terminate simultaneously → single net Close-out Amount. Eliminates cherry-picking.
Master	Section 13 — Governing Law	English law vs New York law — material consequences for insolvency set-off and close-out mechanics
Schedule	Threshold & MTA Elections	Threshold: exposure before VM required. MTA: minimum call size. Both create residual uncollateralised exposure.
Schedule	ATEs & Specified Entities	Additional Termination Events (e.g. rating downgrade, NAV decline). Cross-default extension to affiliates.
CSA	Eligible Collateral Schedule	Asset types, eligible issuers, haircuts per asset. Cash, G10 govvies, agency — each tier with explicit haircut.
CSA	Rehypothecation Rights	Whether receiver may reuse posted collateral. Permitted for VM; prohibited for UMR-segregated IM.

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1992 → 2002 → Current: What Changed and Why

Feature	ISDA 1992	ISDA 2002	Why It Matters
Close-out method	Dual: Market Quotation OR Loss	**Single: Close-out Amount**	Market Quotation failed in illiquid post-Lehman markets — 2002 is self-contained
MQ fallback	Falls back to Loss if quotes unavailable — creates uncertainty	N/A — no fallback needed	66,000 Lehman trades exposed the MQ failure mode
Force Majeure	Not provided — significant gap	Introduced as Termination Event ✓	Addresses impossibility of performance vs unwillingness
Set-off rights	Limited provisions	Expanded and clarified	Broader netting in close-out strengthens recovery
Industry use	Still in use — legacy relationships	Preferred for new dealer-to-dealer docs	Both versions coexist — key to know the differences

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The Lehman Stress Test — September 2008

~66,000 OTC derivative contracts | ~8,000 counterparties | Multiple insolvency jurisdictions

Market Quotation Failure

Dealers refused to quote replacements in illiquid markets. Non-defaulting parties fell back to Loss — the 1992 weakness exposed.

Dispute Explosion

Thousands of bilateral valuation disputes. Complex structured products — model values diverged widely. Resolution took years.

Cherry-Picking Attempts

Lehman administrators initially tested selective enforcement. Courts upheld netting in US, UK, Europe — but the process was slow and costly.

Jurisdictional Fragmentation

US, UK, Germany, Japan — each with separate administrators, different insolvency regimes, adversarial coordination.

Key outcome: Close-out netting held in all major jurisdictions. The 2002 Close-out Amount design was validated. The process was painful — but the framework survived.

Netting: Three Types, One Legal Foundation

Payment Netting

Same-currency, same-date flows netted to a single payment. Section 2(c) ISDA. Reduces settlement risk and operational cost within a single transaction.

Close-out Netting

On default — all trades terminate simultaneously. Single net Close-out Amount replaces all gross obligations. Eliminates cherry-picking. The core ISDA benefit.

Portfolio Netting

Aggregation across all trades in a netting set. All transactions under one Master Agreement → one net exposure for capital (SA-CCR) and close-out.

ENFORCEABILITY RULE

Netting opinions (from local counsel) must confirm enforceability in the counterparty's insolvency jurisdiction. Without a valid opinion, gross exposure applies — dramatically increasing SA-CCR EAD and RWA.

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Worked Example — 5-Trade Portfolio

EUR 6.4M
Gross ITM Exposure
(no netting)

EUR 3.0M
Net Exposure
after Close-out Netting

53%
Credit Exposure
Reduction

~69%
SA-CCR EAD
Reduction

Trade	Notional	MtM (Dealer)	After Netting
IRS-01 (pay fixed, 5Y)	EUR 100M	**+EUR 4.2M**	Included in net — offsets IRS-02
IRS-02 (recv fixed, 3Y)	EUR 75M	**−EUR 2.8M**	Included in net — partially offset by IRS-01/03IRS-03
IRS-03 (pay fixed, 7Y)	EUR 50M	**+EUR 1.9M**	Included in net — offsets
CDS-01 (prot sold, 5Y)	EUR 25M	**−EUR 0.6M**	Included in net — partially offset
FX Forward (6M)	EUR 30M	**+EUR 0.3M**	Included in net — offsets CDS-01
NET POSITION		**+EUR 3.0M**	Single close-out sum

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Variation Margin — Daily MtM Settlement

How VM Works

1. Valuation Agent calculates net MtM of netting set each Valuation Date (daily)
2. If value moved in favour of Party A → Party B posts collateral equal to the change
3. Cash VM earns OIS rate (ESTR/SOFR/SONIA) per CSA terms
4. Threshold and MTA define the band of uncollateralised residual exposure
5. Disputes resolved via IVA polling within agreed tolerance band

Eligible Collateral & Haircuts

Asset	Haircut
Cash (base currency)	**0%**
G10 Govvies ≤2Y	0.5–1%
G10 Govvies 2–10Y	2–4%
G10 Govvies >10Y	4–8%
IG Corporate Bonds	6–12%
Equities	**15–25%**

XVA LINK

Full VM collateralisation → near-zero CVA (daily credit exposure settled) and near-zero FVA (trade funded by VM flows, not dealer balance sheet).

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Initial Margin — Buffer Against MPoR Exposure

Variation Margin (VM)

Settles what HAS already happened
→ Daily MtM movement
→ Eliminates accumulated exposure

Initial Margin (IM)

Buffers against what MIGHT happen
→ Covers MPoR exposure (10 days bilateral)
→ Sized at 99th percentile over MPoR

$$\text{SIMM IM} = \sqrt{\left[\Sigma_i (WS_i^2) + \Sigma_{i \neq j} (\rho_{ij} \times WS_i \times WS_j) \right]}$$

WS_i = weighted sensitivity per risk bucket | ρ_{ij} = intra-bucket correlation | 6 risk classes: IR, CR-Q, CR-NQ, EQ, FX, CM

Context	MPoR Floor	Key Rationale
Bilateral — daily margining	**10 business days**	Detect default → notify → regulatory approval → close-out → replace positions
Centrally cleared	**5 business days**	CCP standardised close-out; position porting to another member
Stress scenario	**20–30 business days**	Lehman-style extended close-out; used in internal stress testing

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Repo — Structure, Mechanics & Types

Securities Seller (Cash Borrower)

DAY 0: Bonds → @ P_0

DAY 0: Cash ← @ P_0

MATURITY: Bonds ← @ same bond

MATURITY: Cash → @ P_1 (P_0 + repo interest)

Cash Lender (Securities Buyer)

MATURITY: Cash → @ $P_1 = P_0 \times (1 + r \times t/360)$

Feature	GC Repo	Special Repo
Collateral	Any eligible bond — lender indifferent	Specific named security — lender requires this exact bond
Rate	Close to OIS — pure funding rate	Below GC — specialness premium for scarcity
Driver	Funding — securities holder needs liquidity	Short-covering — dealer needs specific bond to deliver

FVA LINK

GC repo rates form the lower bound of the dealer's secured funding curve. The spread between GC repo and unsecured CP/CD is the primary driver of FVA on uncollateralised derivative portfolios.

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Collateral Transformation & Haircut Procyclicality

The Collateral Upgrade Trade

1. Client holds IG corporate bonds — ineligible as CCP margin
2. Client approaches transformation provider (dealer/custodian)
3. Provider lends eligible govt bonds in exchange for IG bonds + transformation fee
4. Client posts govt bonds as CCP IM — margin obligation met
5. At maturity: trade unwinds, both sides reclaim original assets

Transformation fee: 10–50 bps p.a. — enters MVA as all-in cost of eligible IM collateral

Haircut Procyclicality — A Systemic Risk

Good Times

Haircuts compress as competition drives margins down. Near-zero haircuts on IG bonds common.

Stress Onset

Haircuts spike simultaneously. All participants margin-call at once — exactly when liquidity most scarce.

Crisis

Rising haircuts force liquidations. Liquidation depresses prices. Prices trigger further haircut increases — collateral spiral.

FSB minimum haircut floors (BCBS-IOSCO / SFTR) were designed to prevent compression in good times.

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Archegos (2021) — A Modern Collateral Management Lesson

Family office | Leveraged TRS exposure across multiple dealers | No cross-dealer netting | Limited or no IM

No Cross-Dealer Netting Set

Each dealer saw only its own bilateral exposure. No visibility into Archegos's aggregate leverage across all counterparties simultaneously.

No IM / Insufficient IM

Archegos was below UMR thresholds (Phase 5/6 activating in 2021–22). Several dealers negotiated minimal or zero IM — leaving large uncollateralised exposures.

Margin Call Collateral Spiral

Stock price decline → simultaneous margin calls from all dealers → Archegos forced to liquidate → further price decline → further calls. Classic collateral spiral.

Information Asymmetry

The bilateral framework has no mechanism for dealers to share aggregate exposure information. Each dealer optimised bilaterally; system-level concentration was invisible.

~USD 5.5B	~USD 2.9B	>USD 10B
Credit Suisse	Nomura	Industry Total

Regulatory Landscape — EMIR, Dodd-Frank, MAS, HKMA

Regime	Jurisdiction	Clearing Obligation	Margin Rules	Reporting
EMIR	EU/UK	IRS/CDS index — LCH, Eurex	UMR VM+IM — BCBS-IOSCO aligned	T+1 to DTCC/REGIS-TR
Dodd-Frank	US	SEF trading + DCO clearing	CFTC/PR IM rules — SIMM accepted	Real-time to CFTC SDR
MAS SFA	Singapore	SGD IRS + USD/EUR IRS above threshold	SFA 04-N12 — Phase 5/6 extended	SFA 04-N05 — MAS TR
HKMA SPM	Hong Kong	HKD + USD/EUR IRS; LCH recognised	CR-G-14 — international timeline	HKTR framework

APAC NOTE

MAS and HKMA have both confirmed cross-border equivalence with EMIR and CFTC frameworks, enabling entities to avoid duplicative compliance where bilateral recognition agreements are in place.

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Updated XVA Waterfall — Where Collateral Sits

CVA

Counterparty credit risk. Reduced by VM (daily settlement eliminates build-up). Residual driven by Threshold + MTA gap.

DVA

Own credit adjustment — bilateral. Not affected directly by collateral regime.

FVA

Funding cost of uncollateralised exposures. VM-collateralised trades → near-zero FVA. Residual driven by Threshold/MTA uncollateralised band.

MVA

Funding cost of posting IM. Driven by SIMM IM amount × funding spread. Collateral transformation fee adds to all-in MVA.

ColVA

Cheapest-to-deliver optionality in multi-currency CSAs. COVERED IN MODULE 6.

KVA

Cost of regulatory capital. COVERED IN MODULE 6.

Module 5 — Key Takeaways

01 The ISDA hierarchy: Master + Schedule (inseparable, can stand alone) → CSA (optional, cannot exist without Master) → Confirmations. Know the dependency logic.

02 Close-out netting is the core ISDA benefit — eliminating cherry-picking and enabling net SA-CCR EAD for capital. It only works if netting opinions confirm enforceability.

03 1992 vs 2002: Market Quotation failed in illiquid Lehman markets. The 2002 single Close-out Amount is the correct framework — but both versions remain in use.

04 VM eliminates current MtM exposure daily. IM buffers against MPoR exposure (10 days bilateral). UMR mandates segregation of IM — prohibiting rehypothecation.

05 Repo is the dealer's funding infrastructure. GC repo forms the secured funding curve lower bound. Special repo reflects bond scarcity. Both feed FVA and MVA.

06 Collateral transformation (IG bonds → govt bonds) enables CCP margin posting — at a transformation fee that enters MVA. Haircut procyclicality amplifies stress.

Coming Next — Module 6

MODULE 6

KVA, ColVA & Capital Costs — Completing the XVA Waterfall

ColVA	Full derivation — cheapest-to-deliver optionality in multi-currency CSAs. Worked example on EUR 100M IRS.
KVA	Capital Valuation Adjustment — cost of regulatory capital. CET1 hurdle rate, SA-CCR EAD/RWA, stress capital buffer, G-SIB interaction.
XVA Completion	Updated waterfall with all six adjustments. Desk-level case studies across three collateral regimes. P&L attribution framework.

Series: Basel I–IV → FRTB → CVA/DVA → FVA/MVA → ISDA, Netting & Collateral → KVA, ColVA & Capital Costs → Practitioner's Dictionary